AI Data Science Copilot

Automated Dataset Intelligence, Health, & Model Report

Domain:	Healthcare
Dataset Type:	Tabular
Problem Type:	Binary Classification
Target Column:	Outcome
Total Rows:	768
Total Columns:	9

Generated by Antigravity AI Data Science Copilot Engine

1. Dataset Health Assessment

Overall Dataset Health Score: 100/100

The health score evaluates data quality dimensions including duplicates, outliers, missingness, and collinearity:

Metric	Value
Missing Cells Percentage	0.0%
Duplicate Rows Count	0 (0.0%)
Outliers Percentage	0.0% (0 values)
Memory Usage	0.05 MB

2. Cleaning Operations Log

The following cleaning steps were applied to the raw dataset:

- `Capped 4 outliers in column 'Pregnancies' at IQR bounds [-6.50, 13.50].`
- `Capped 5 outliers in column 'Glucose' at IQR bounds [37.12, 202.12].`
- `Capped 45 outliers in column 'BloodPressure' at IQR bounds [35.00, 107.00].`
- `Capped 1 outliers in column 'SkinThickness' at IQR bounds [-48.00, 80.00].`
- `Capped 34 outliers in column 'Insulin' at IQR bounds [-190.88, 318.12].`
- `Capped 19 outliers in column 'BMI' at IQR bounds [13.35, 50.55].`
- `Capped 29 outliers in column 'DiabetesPedigreeFunction' at IQR bounds [-0.33, 1.20].`
- `Capped 9 outliers in column 'Age' at IQR bounds [-1.50, 66.50].`

3. Bias & Fairness Analysis

Evaluated sensitive dimension: **'Age'** (Privileged: '22.0')

Disparate Impact Ratio: **2.42** (Status: **BIASED**)

Statistical Explanation: In the historical labels, the unprivileged group has a favorable outcome rate of **36.9%**, compared to **15.3%** for the privileged group. Since the Disparate Impact ratio (**2.42**) is outside the standard 80-120% fairness boundary (0.80 - 1.25), there is statistically significant evidence of potential **historical/underrepresentation bias** in this dataset.

4. Model Performance Comparison

Algorithm	Training Time	Inference Time	ACCURACY	PRECISION	RECALL	F1	ROC_AUC
Logistic Regression	0.042s	0.003s	0.7532	0.6545	0.6545	0.6545	0.8145
Random Forest	0.271s	0.014s	0.7208	0.6071	0.6182	0.6126	0.8096
XGBoost	0.342s	0.008s	0.7143	0.5902	0.6545	0.6207	0.7607

5. AI Business Insights & Strategy

Executive Business Insights Report

Overview of Dataset Analysis

The provided dataset consists of 768 rows and 9 columns, indicating it is relatively small but comprehensive in its scope. The dataset has been classified as balanced, suggesting that the classes are evenly distributed without any significant skewness. This characteristic is beneficial for many machine learning algorithms which rely on a well-balanced distribution to avoid biased models.

The memory footprint of approximately 0.05 MB suggests that this dataset can be easily managed and analyzed within standard computing resources. Given its size, it's ideal for rapid prototyping, exploratory data analysis (EDA), and initial model development without the need for extensive storage or computational power.

Key Observations

1. **Data Distribution:** - The balanced nature of the dataset ensures that no single class dominates the results, which is crucial for maintaining unbiased models and ensuring robust generalization across different classes. 2. **Column Characteristics:** - Each column represents a distinct feature or variable in the dataset. Understanding these features will be pivotal to uncovering meaningful insights.

3. **Memory Efficiency:** - The small memory footprint indicates that the data is efficiently stored, facilitating quick analysis and model training without compromising on performance.

Business Insights

1. **Balanced Data Implications:** - Given the balanced nature of the dataset, it's essential to ensure that all features are equally represented across different classes. This balance can be maintained by conducting thorough feature engineering if necessary or by ensuring data preprocessing steps like normalization and encoding do not introduce any bias.

2. **Feature Importance Analysis:** - Analyzing which columns (features) have the most significant impact on the target variable will provide valuable insights into what factors are driving business outcomes. This can guide decision-making processes, such as prioritizing certain areas for improvement or investment.

3. **Data Quality Checks:** - Since the dataset is small and balanced, it's crucial to perform rigorous data quality checks. Look out for any anomalies in the data, missing values, outliers, and ensure that all categorical variables are properly encoded. Addressing these issues early can prevent model performance degradation due to poor data integrity.

Recommendations

1. **Feature Engineering:** - Develop a comprehensive feature set by exploring potential interactions between existing features or creating new ones based on domain knowledge. This step is critical for enhancing the

predictive power of models and ensuring that all relevant information is captured.

2. **Model Selection & Validation:** - Based on the insights from EDA, select appropriate machine learning algorithms suitable for your business problem. Validate these models using cross-validation techniques to ensure they generalize well across different subsets of data.

3. **Iterative Improvement:** - Implement an iterative improvement cycle where initial model performance is evaluated against real-world outcomes. Use this feedback loop to refine the feature set, adjust hyperparameters, and potentially retrain models with additional data or enhanced features.

By adhering to these recommendations, businesses can leverage their datasets more effectively, leading to informed decision-making that drives strategic growth and operational efficiency.